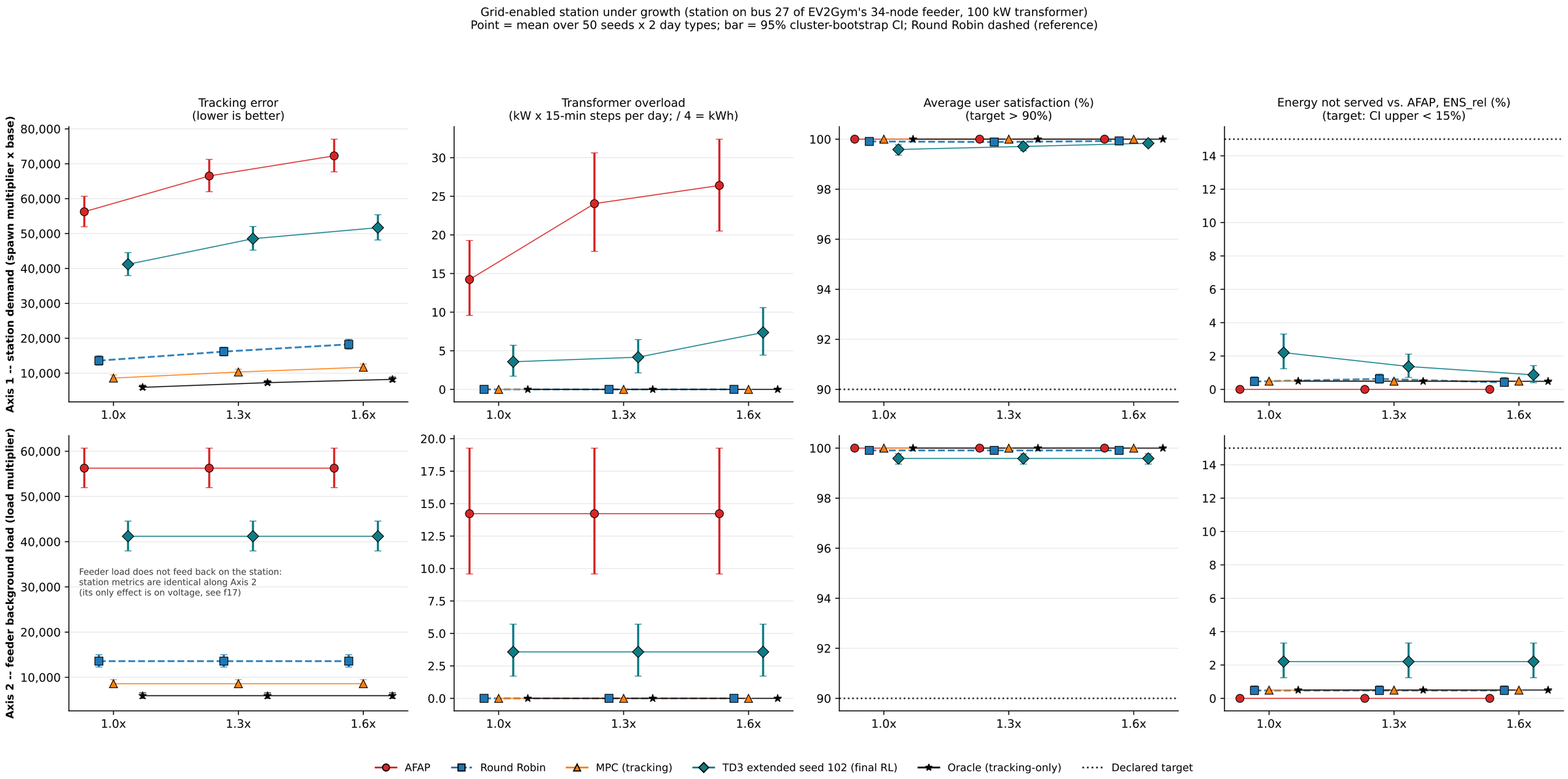
Grid-enabled station under growth (station on bus 27 of EV2Gym's 34-node feeder, 100 kW transformer)
Point = mean over 50 seeds x 2 day types; bar = 95% cluster-bootstrap CI; Round Robin dashed (reference)
Tracking error
(lower is better)
Transformer overload
(kW x 15-min steps per day; / 4 = kWh)
Average user satisfaction (%)
(target > 90%)
Energy not served vs. AFAP, ENS_rel (%)
(target: CI upper < 15%)
Axis 1 -- station demand (spawn multiplier x base)
Axis 2 -- feeder background load (load multiplier)
1.0x
1.3x
1.6x
Feeder load does not feed back on the station:
station metrics are identical along Axis 2
(its only effect is on voltage, see f17)
AFAP
Round Robin
MPC (tracking)
TD3 extended seed 102 (final RL)
Oracle (tracking-only)
Declared target